



**COMSATS University Islamabad,
Abbottabad Campus**

Project Proposal

(SCOPE DOCUMENT)

for

INTELI-REHAB

A Smart Wearable Rehabilitation System

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Project Category: (Select all the major domains of proposed project)

☐ A -Desktop Application/Information System	☐ B -Web Application/Web Application based Information System
☐ C - Problem Solving and Artificial Intelligence	☐ D -Simulation and Modelling
☐ E - Smartphone Application	☐ F - Smartphone Game
☐ G - Networks	☐ H - Image Processing
☐ Other (specify category) _____	

Abstract

Inteli-Rehab is a clinic-integrated mobile rehabilitation platform designed to bridge the gap between in-clinic physiotherapy and effective home recovery for upper-arm musculoskeletal injuries. Rather than replacing clinical care, the system extends it: patients register through their hospital or clinic, select an appointed physiotherapist, and receive a personalized rehabilitation protocol drawn from a curated clinical exercise database, but one built and maintained by the physiotherapy staff for their specific patient population. Each protocol targets the affected joint with precise goals, such as achieving full arm flexion through progressive repetitions, while the physiotherapist retains full control over exercise selection, intensity, and range-of-motion targets.

The patient experience centers on a wearable upper-arm band equipped with dual IMU and EMG sensors. After Bluetooth pairing, the system captures the patient's neutral joint position and resting muscle baseline during a guided calibration sequence. During live sessions, a real-time 3D digital twin of the patient's arm renders movement with color-coded muscle activation, shifting from cool blues for low engagement to warm reds for peak contraction, while an onboard machine learning model continuously monitors for abnormal movement patterns and compensatory behaviours. Deviations trigger immediate visual alerts on the twin and voice-guided corrections, ensuring safe, supervised-quality form even when the patient exercises alone. Session completion feeds into a motivation system of streaks, recovery levels, and achievement badges tied to tangible progress toward the clinician-set 100% range-of-motion goal. All performance data syncs to a secure cloud backend, where the physiotherapist reviews objective metrics, adjusts protocols, and approves progression, transforming intermittent clinic visits into a continuous, data-informed recovery partnership.

Introduction

Inteli-Rehab is a wearable-based rehabilitation system for upper-limb injuries that integrates IMU and EMG sensors, mobile computing, and AI-driven analysis to enable continuous monitoring beyond clinical settings. The system addresses the lack of real-time feedback in home-based physiotherapy, where patients typically perform exercises without accurate guidance on movement quality or muscle activity.

The wearable unit streams sensor data to a mobile application via Bluetooth Low Energy, where sensor fusion and signal processing are used to estimate joint motion and muscle activation. A real-time digital twin visualizes limb movement, while machine learning models detect abnormal patterns and fatigue indicators to support corrective feedback.

All session data is synchronized to a cloud platform, allowing physiotherapists to monitor patient progress and adjust rehabilitation plans remotely. This creates a connected rehabilitation workflow linking patient activity, AI analysis, and clinical supervision.

Problem Statement

Upper-limb rehabilitation patients often perform home exercises without real-time clinical supervision, leading to incorrect movement patterns, poor exercise quality, and slow recovery. Physiotherapists also lack continuous, objective data to monitor patient progress between sessions, relying mostly on periodic visits and self-reports.

Existing rehabilitation solutions either focus on basic motion tracking or camera-based systems, but they do not provide integrated monitoring of both joint movement and muscle activity in real time. This limits their ability to detect compensatory movements, fatigue, and improper form during rehabilitation exercises.

Inteli-Rehab addresses this gap by combining wearable IMU and EMG sensors with AI-based movement analysis and digital twin visualization, enabling accurate real-time feedback for patients and remote monitoring for physiotherapists.

Problem Solution for Proposed System

Inteli-Rehab solves identified rehabilitation challenges through a multi-layered intelligent system architecture integrating wearable hardware, real-time data processing, AI-based analysis, and intuitive visualization.

1. **Wearable Sensing & Data Acquisition:** Dual IMU sensors and dual-channel surface EMG sensors worn on the affected upper limb continuously stream motion and muscle data to a mobile application via Bluetooth Low Energy (BLE).
2. **Sensor Fusion & Signal Processing:** IMU data is processed using Madgwick filter or Extended Kalman Filter (EKF) to compute stable joint angles and range of motion (ROM)

in real-time. EMG signals undergo filtering and feature extraction to determine muscle activation levels and detect fatigue trends through median frequency analysis.

3. **3D Digital Twin & Real-Time Feedback:** Processed kinematic and physiological data feeds into a 3D digital twin that renders the patient's limb movement in real-time and highlights deviations from correct exercise form using color-coded visual cues.
4. **AI Anomaly Detection:** AI models trained on labelled movement datasets analyse fused sensor data to automatically detect abnormal movement patterns, compensatory behaviours, and early fatigue indicators, triggering real-time alerts to the patient.
5. **Gamified Patient Interface:** The mobile application presents exercise instructions, live feedback, repetition counting, and performance scoring through a gamified interface including progress badges and recovery levels to sustain patient motivation and adherence
6. **Remote Clinical Oversight:** All session data is securely synchronized to a cloud backend powered by Supabase, where physiotherapists access a dedicated dashboard to review patient progress, approve exercise progression, adjust rehabilitation protocols, and generate performance reports.
7. **Personalized Calibration:** Patient-specific calibration during initial setup accounts for individual anatomical differences, making feedback personalized and clinically meaningful.

Related System Analysis/Literature Review

Several digital rehabilitation systems have been developed to support remote physiotherapy, each addressing specific aspects of motion tracking and patient engagement.

SWORD Health utilizes wearable IMU sensors combined with remote therapist supervision to monitor patient exercises. While effective in capturing movement data, the system does not incorporate muscle-level sensing such as EMG, limiting its ability to analyse fatigue or muscle imbalance. It also lacks immersive real-time biomechanical visualization.

Kaia Health relies on smartphone camera-based pose estimation to guide rehabilitation exercises. Although accessible without additional hardware, its accuracy is affected by lighting conditions, camera positioning, and occlusion. Furthermore, it does not provide physiological muscle activity analysis or therapist-side monitoring tools.

Hinge Health provides structured digital physiotherapy programs with wearable motion sensors and remote coaching. However, its analytical capabilities are primarily focused on adherence tracking rather than detailed biomechanical correction or AI-based anomaly detection. Real-time visualization of limb kinematics is also limited.

In comparison, Inteli-Rehab integrates IMU and EMG sensing with AI-driven anomaly detection and a real-time digital twin. This combination enables simultaneous analysis of movement mechanics and muscle activity, offering a more comprehensive and clinically informative rehabilitation approach.

Table 1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
SWORD Health	No EMG muscle assessment; lacks 3D digital twin visualization; limited patient engagement features.	Integrates dual IMU + EMG sensors for comprehensive motion and muscle monitoring; provides real-time 3D digital twin with deviation highlighting; includes gamification for enhanced engagement.
Kaia Health	Camera-dependent pose estimation with accuracy limitations; no muscle activation or fatigue assessment; no physiotherapist remote monitoring dashboard.	Uses wearable IMU sensors for precise joint angle measurement independent of camera; includes EMG fatigue detection; provides cloud-based physiotherapist dashboard for remote oversight.
Hinge Health	Lacks AI-driven compensatory movement detection; no real-time biomechanical digital twin; focuses on adherence rather than form correction.	Employs autoencoder/LSTM-based AI models for abnormal and compensatory movement detection; renders real-time 3D kinematic digital twin; provides form correction alerts and adaptive exercise recommendations.

Advantages/Benefits of Proposed System

1. Objective and Continuous Monitoring

Unlike subjective self-assessment or periodic clinical visits, Inteli-Rehab provides continuous, data-driven feedback on exercise form, joint angles, and muscle activation, ensuring rehabilitation quality is maintained throughout every session.

2. Early Detection of Compensatory Movements and Fatigue

The AI-based analysis detects abnormal patterns such as trunk lean, momentum compensation, and muscle fatigue in real-time, preventing incorrect exercise execution that could lead to re-injury or delayed recovery.

3. Remote Physiotherapist Oversight

The cloud-based dashboard enables physiotherapists to monitor multiple patients remotely, review detailed performance metrics, adjust exercise protocols, and provide timely interventions without requiring in-person appointments.

4. Enhanced Patient Engagement Through Gamification

Repetition counting, achievement badges, and progression levels transform rehabilitation from a repetitive chore into an interactive and motivating experience, significantly improving patient adherence rates.

5. Personalized and Adaptive Rehabilitation

Patient-specific sensor calibration and adaptive exercise recommendation modules ensure that feedback and exercise plans are tailored to individual anatomical characteristics and recovery progress.

6. Scalable and Accessible Healthcare Deliver

The system reduces dependency on clinical infrastructure and physiotherapist availability, making quality rehabilitation accessible to patients in remote or under-resourced areas at a fraction of traditional therapy costs.

7. Clinically Relevant Data Security and Reporting

Secure authentication, encrypted data storage, and detailed progress reports ensure patient data privacy while providing physiotherapists with actionable clinical insights for evidence-based treatment decisions.

Scope

In Scope

1. Wearable hardware prototype with dual IMU (MPU6500) and dual-channel EMG sensors, BLE-enabled microcontroller
2. Flutter-based Android mobile application for BLE connectivity, sensor data reception, calibration, and exercise session management
3. Sensor fusion for joint angles and ROM computation
4. EMG signal processing: filtering, feature extraction, and fatigue trend analysis
5. 3D digital twin visualization (Unity/Three.js) with real-time movement rendering and color-coded deviation feedback
6. AI anomaly detection for abnormal patterns, compensatory behaviours, and early fatigue alerts
7. Gamification: repetition counting, performance scoring, badges, and achievement levels
8. Supabase cloud backend for authentication, encrypted storage, session sync, and multi-device support
9. Web based physiotherapist dashboard for remote monitoring, exercise approval, protocol adjustments, and report generation and a clinic web panel to appoint physiotherapists
10. Upper-limb musculoskeletal rehabilitation exercises under physiotherapist supervision

Out of Scope

1. Live video-based pose estimation systems
2. Clinical diagnostic capabilities beyond exercise guidance
3. Integration with hospital electronic medical record (EMR) systems
4. Physics-based biomechanical force simulations

5. Medical prescriptions or replacement of professional clinical diagnosis

Modules

Module 1: Clinic and Patient Onboarding

When someone opens the app for the first time, they pick their hospital or clinic from a list, then choose a physiotherapist who works there. They tell the system what happened to their upper arm, maybe they had surgery, tore a muscle, or broke a bone near the elbow. Their chosen physiotherapist then builds a personal exercise plan for them using the clinic's own collection of trusted workouts, setting clear goals like "bend your arm 5 times, twice a day" and working toward full movement again. The app makes sure only real patients, and real physiotherapists can log in, keeping everyone's information safe.

Module 2: Clinical Exercise Database

The database stores all clinic-approved exercise templates with descriptions and instructions. Physiotherapists select exercises from this database and assign personalized reps, sets, and session frequency for each patient. Patients only see exercises their own therapist has configured and approved.

Module 3: Wearable Sensor and Bluetooth Connection

The armband contains IMU and EMG sensors that track arm movement and muscle activity, sending data wirelessly to the mobile app via Bluetooth Low Energy. The app monitors connection quality, provides placement guidance, and auto-reconnects if signal drops. This module forms the data foundation for all real-time feedback and analysis features.

Module 4: Biometric Baseline Establishment

The app records each patient's unique resting arm position and relaxed muscle baseline during a short setup routine. These personal measurements ensure all future feedback is tailored to the individual's body. Patients recalibrate if the band shifts or after extended use to maintain accurate readings.

Module 5: Movement Tracking and Progress Measurement

The module processes sensor data to calculate real-time shoulder and elbow angles, comparing them against the physiotherapist's targets. It auto-counts repetitions, detects rest periods, and tracks range-of-motion progress over time. Simple charts show patients and therapists whether movement is improving toward full recovery.

Module 6: Live Digital Twin Visualization

The module renders a real-time 3D model of the patient's arm that mirrors every movement during exercise. Muscle activation displays through color coding, blue for low, green for moderate, orange for high, and red for limit or incorrect form. Deviations like compensatory shoulder movement trigger yellow or red alerts, while patients can rotate, zoom, or replay in slow motion for clear visual feedback.

Module 7: Movement Quality Assessment and Corrective Alert System

The module uses machine learning models trained on physiotherapist-approved movement data to detect incorrect form in real time. It identifies compensatory patterns like shoulder shrugging, elbow flaring, and momentum swinging by analyzing both motion and muscle signals together. When anomalies are detected, it triggers instant visual and audio alerts on the 3D model while logging all warnings for therapist review.

Module 8: Fatigue Monitoring and Safety Intervention

The module monitors EMG signal changes to detect muscle fatigue using each patient's personal baseline from calibration. It delivers three-stage warnings—gentle screen hint, spoken reminder, and session pause suggestion—as fatigue increases. All fatigue data is saved and visualized for physiotherapists to adjust exercise intensity and prevent overuse injuries.

Module 9: Physiotherapist Control Center and Cloud Synchronization

The module provides physiotherapists and clinic with a secure web-based dashboard where the clinic panel is limited to adding appointed physiotherapists. The physiotherapist's panel allows monitoring patient progress, adjusting exercise plans, and generating PDF reports remotely. An offline-first sync engine ensures no data loss during connectivity drops, with automatic upload when connection resumes. This transforms intermittent clinic visits into continuous remote oversight for uninterrupted recovery.

System Limitations/Constraints

1. Hardware Cost and Availability:

The cost and commercial availability of quality EMG sensors and IMU modules may limit the initial prototype deployment and scalability of the wearable hardware component, potentially requiring alternative sourcing strategies or simplified sensor configurations.

2. BLE Connectivity and Latency:

Bluetooth Low Energy communication between the wearable sensors and mobile device introduces inherent latency and is susceptible to interference, connection drops, and range limitations, which may affect real-time feedback responsiveness during dynamic exercises.

3. Mobile Device Performance:

Real-time 3D digital twin rendering and AI model inference on mobile devices may be constrained by processing power, memory, and battery life, particularly on mid-range or older Android smartphones, requiring optimization trade-offs between visualization quality and performance.

4. Data Security and Privacy Compliance:

Handling sensitive patient health data requires strict adherence to data protection regulations such as HIPAA or GDPR, necessitating robust encryption, secure authentication, access controls, and audit mechanisms that add complexity to the backend architecture and deployment.

Software Process Methodology

The Inteli-Rehab project will follow the Incremental Development Model, where the system is built step by step in clearly defined phases, each delivering a working part that gradually combines into the complete product. This approach is chosen because Inteli-Rehab is a complex system spanning wearable sensors, signal processing, mobile application development, artificial intelligence models, and cloud-based monitoring, all of which need to be developed and tested sequentially rather than simultaneously. The development will be divided into major milestones aligned with university evaluation stages (10%, 30%, 60%, and 100%), where new features are added atop previous work: early stages focus on basic application and sensor connection, subsequent stages add motion tracking and calibration, later stages integrate AI-based analysis, and the final stage completes the system with visualization, dashboard, and full integration. This model suits Inteli-Rehab because it reduces complexity by breaking the project into manageable

parts, ensures a working system at every stage, allows the team to focus on completing one feature set before advancing, and helps meet strict academic deadlines while maintaining stability and proper hardware-software integration.

Tools and Technologies

The development tools and core technologies utilized in this project are summarized in the following table:

Table 2 Tools and Technologies for Proposed Project

Tools And Technologies	Tools	Version	Rationale
	Android Studio	2023.1.1	IDE for Flutter mobile development
	VS Code	1.85	IDE for Python AI/ML development and backend
	Unity Editor	2022.3 LTS	3D Digital Twin visualization development
	PostgreSQL	15	Relational Database Management System
	Supabase	Latest	Cloud backend platform with authentication and realtime APIs
	Figma	Latest	UI/UX design and mockups creation
	Draw.io	Latest	System architecture and flowchart diagrams
	MS Word	2023	Documentation
	MS PowerPoint	2023	Presentation
	GitHub	Latest	Version control and collaboration
	Technology	Version	Rationale
	Dart (Flutter)	3.11.0	Cross-platform mobile application development
	React.js	8.x	Used for building the web-based physiotherapist and clinic dashboards with reusable UI components and efficient state management
	Python	3.14.2	AI/ML model development and signal processing
	TensorFlow/PyTorch	2.x/2.x	Deep learning frameworks for anomaly detection models

	C# (Unity)	2022.3	3D digital twin visualization scripting
	Three.js	Latest	Alternative web-based 3D visualization
	SQL	PostgreSQL 15	Database query language
	BLE Protocol	5.0	Wireless communication with wearable sensor
	REST API	HTTP/2	Client-server communication
	JWT	Latest	Secure authentication token standard

Project Stakeholders and Roles

The project stakeholders and their respective roles and responsibilities are outlined in the following table:

Table 3 Project Stakeholders for Proposed Project

Project Sponsor	COMSATS University, Islamabad
Stakeholder	Roles and Responsibilities
Students: Tehreem Zaheer, Kashmala Zeb and Ahmed Irshad	Responsible for the design, development, implementation, and testing of the “Inteli-Rehab” Ensure all project requirements are fulfilled, and proper documentation is maintained.
Project Supervisor: Dr. Ahmed Khan	Provides continuous guidance, feedback, and supervision throughout the project lifecycle. Ensures that the project aligns with academic and technical standards.
Final Year Project Committee	Evaluates the project in terms of functionality, innovation, technical depth, documentation, and presentation. Provides feedback and final grading.

Team Members Individual Tasks/Work Division

Team members and their respective modules are listed in the following table:

Table 4 Team Member Work Division for Proposed Project

Student Name	Student Registration Number	Responsibility/ Modules
Tehreem Zaheer	Sp23-BCS-131	Clinic and Patient Onboarding (Module 1), Clinical Exercise Database (Module 2), Wearable Sensor and Bluetooth Connection (Module 3), Biometric Baseline Establishment (Module 4), Mobile application development, BLE connectivity implementation, UI/UX design, and project documentation
Kashmala Zeb	Sp23-BSE-048	Movement Tracking and Progress Measurement (Module 5), Live Digital Twin Visualization (Module 6), Movement Quality Assessment and Corrective Alert System (Module 7).
Ahmed Irshad	Sp23-BSE-045	Fatigue Monitoring and Safety Intervention (Module 8), Physiotherapist Control Center and Cloud Synchronization (Module 9), Machine learning model training, signal processing, and 3D visualization development.

Data Gathering Approach

The requirements and data for Inteli-Rehab will be gathered from the EMG Elbow Dataset and PhysioNet, specialized public dataset focused on upper-arm muscle activity during elbow movements including flexion, extension, supination, and pronation. This dataset provides directly relevant EMG signals from the biceps brachii and triceps brachii muscles, enabling accurate AI model training for fatigue detection, muscle activation analysis, and movement pattern recognition specific to upper-limb rehabilitation. Supplementary requirements will be refined through structured consultations with licensed physiotherapists and patients to ensure clinical alignment.

Concepts

Concept 1: Sensor Fusion

Sensor fusion is the process of combining data from multiple sensors to produce more accurate, reliable, and comprehensive information than individual sensors could provide alone. In Inteli-Rehab, sensor fusion algorithms such as the Madgwick filter and Extended Kalman Filter integrate accelerometer, gyroscope, and magnetometer readings from dual IMUs to compute stable joint angles and orientations, compensating for the drift and noise inherent in individual sensor components.

Concept 2: Electromyography (EMG) Signal Processing

Electromyography signal processing involves the acquisition, filtering, and analysis of electrical signals generated by muscle contractions. In this project, raw EMG signals from surface electrodes undergo bandpass filtering to remove motion artifacts and noise, followed by rectification, smoothing, and feature extraction to calculate metrics such as root mean square amplitude and median frequency, which are used to assess muscle activation levels and detect fatigue trends during rehabilitation exercises.

Concept 3: Digital Twin

A digital twin is a virtual representation of a physical object or system that is updated with real-time data to mirror its state and behavior. Inteli-Rehab employs a kinematic digital twin that renders a 3D anatomical model of the patient's limb, updating its pose in real-time based on processed IMU data and highlighting deviations from correct exercise form, thereby providing intuitive visual feedback that enhances patient self-correction capabilities.

Concept 4: AI Models

Autoencoders are unsupervised neural networks that learn efficient data representations by compressing input data into a latent space and reconstructing it, with reconstruction errors indicating anomalies. Long Short-Term Memory networks are recurrent neural networks capable of learning long-term dependencies in sequential data. In Inteli-Rehab, autoencoders detect abnormal movement patterns through reconstruction error analysis, while LSTM networks model the temporal dynamics of exercise sequences for compensatory movement classification and fatigue prediction.

Concept 5: Bluetooth Low Energy (BLE) Communication

Bluetooth Low Energy is a wireless communication protocol designed for short-range transmission with minimal power consumption, making it ideal for wearable devices. Inteli-Rehab utilizes BLE 5.0 to establish a low-latency, energy-efficient connection between the wearable sensor microcontroller and the mobile application, enabling continuous streaming of IMU and EMG data while preserving battery life for extended rehabilitation sessions.

Concept 7: Human-Computer Interaction (HCI)

Human-Computer Interaction is the study and design of how people interact with technology, focusing on usability, accessibility, and user experience. In Inteli-Rehab, HCI principles ensure the patient app is approachable for older adults and non-technical users through large touch targets, clear visual hierarchy, voice alerts for hands-free feedback, and color-blind friendly digital twin palettes. The physiotherapist dashboard applies HCI through intuitive data visualization,

presenting complex metrics as actionable charts rather than raw numbers. By grounding design in HCI research, Intelli-Rehab ensures powerful technology remains genuinely helpful to all users.

Gantt chart

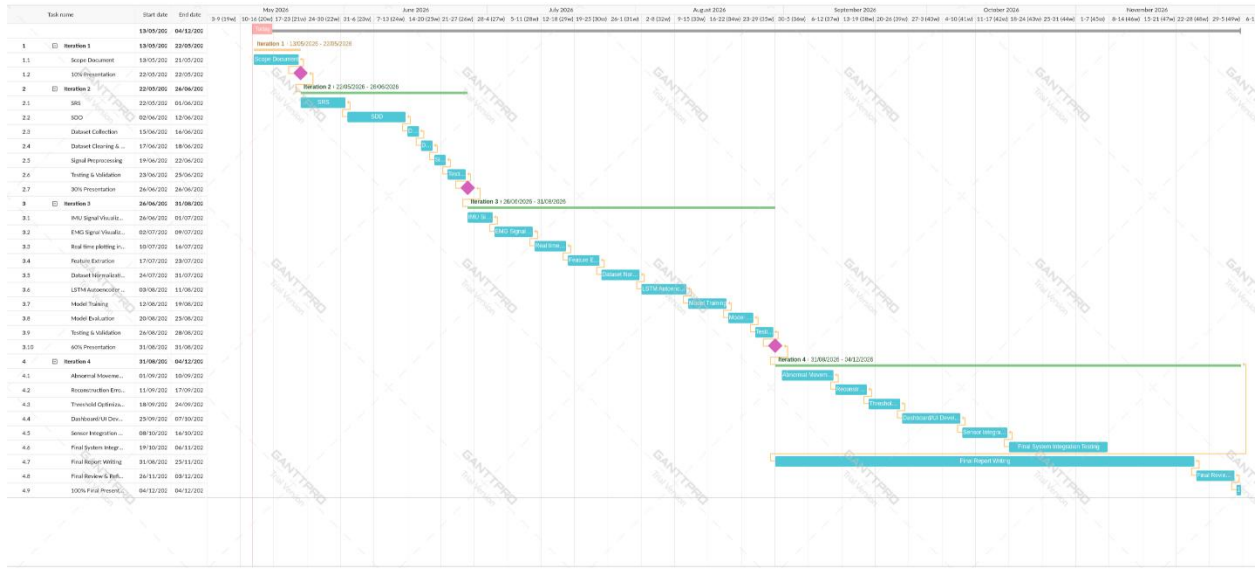


Figure 1 Gantt Chart of Intelli-Rehab

Mockups

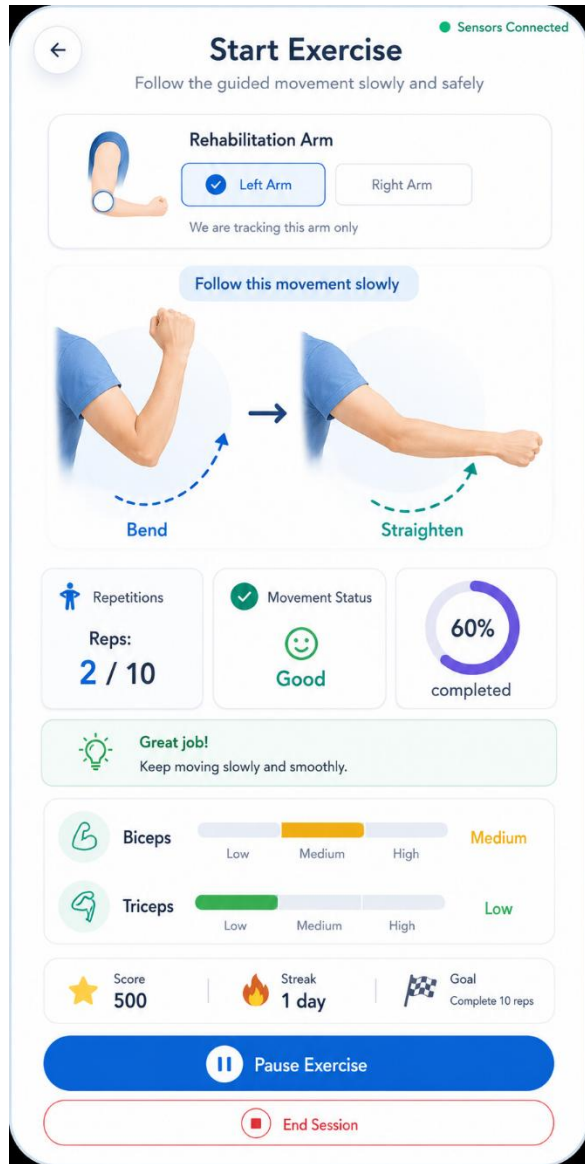


Figure 3 Live Exercise Session Tracking

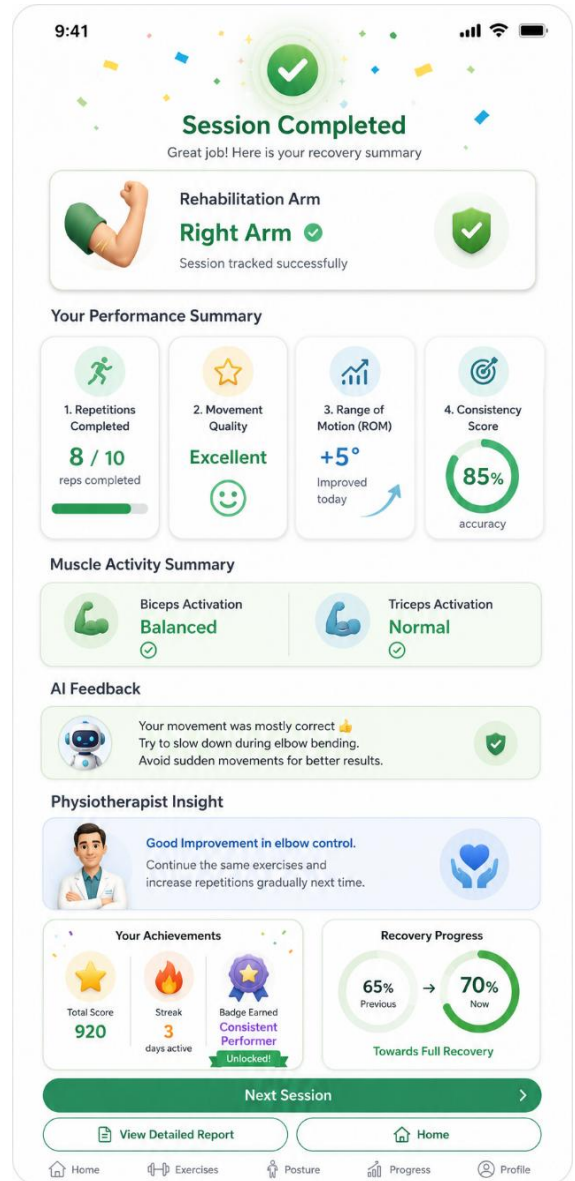


Figure 2 Exercise Summary

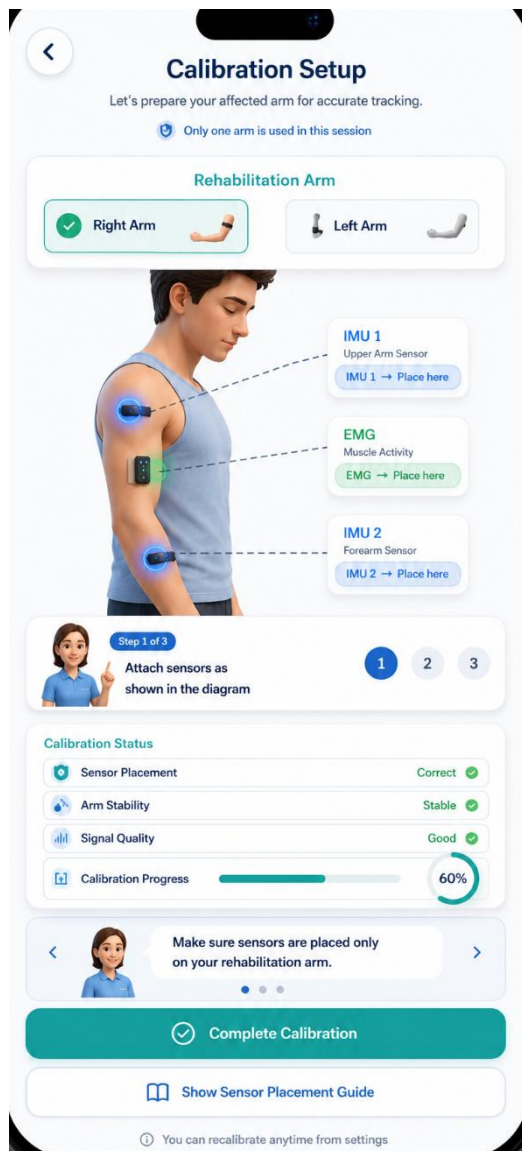


Figure 5 Calibration Setup Page

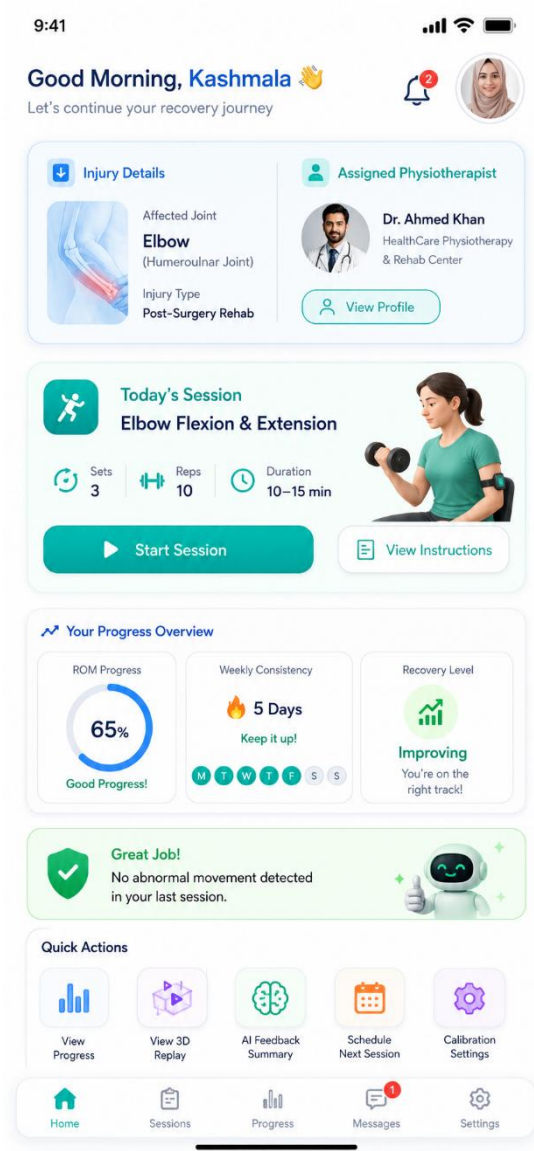


Figure 4 Patient Dashboard

Conclusion

Inteli-Rehab is an intelligent rehabilitation system designed to help patients recover from musculoskeletal injuries using wearable sensors, artificial intelligence, and 3D visualization. The system uses dual IMU and EMG sensors to track movement and muscle activity, while AI-based analysis helps detect incorrect movements and provides real-time feedback. A 3D digital twin is used to visually show patient movement in an easy-to-understand way, and physiotherapists can monitor patient progress remotely through the system. The platform also includes gamification features to motivate patients and improve exercise consistency. The system is developed using the Incremental Development Model, where the project is built step by step in phases (10%, 30%, 60%, and 100%), and each phase adds new features to the

previous version until the full system is completed. This project aims to improve rehabilitation outcomes by making exercises more accurate, engaging, and remotely monitorable. It also helps the development team gain practical experience in hardware integration, mobile app development, artificial intelligence, and cloud-based systems.

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